

```
experimenting"
Backtrace:
 1. testthat::expect_match(quote, "science")
 2. testthat::expect_match(...)
Error: Test failed

> test_that ("expect match", {
  expect_match(quote, "science", ignore.case = TRUE)
})
Test passed
```

You can pass an *ignore.case=TRUE* third argument option to the *expect_match()* function for a case insensitive match.

The *expect_message*, *expect_warning*, *expect_error*, and *expect_output* functions are a variant of *expect_match* that examine the output. Examples of their usage are given below:

```
> test_that ("expect message", {
  expect_message(message("2021-02-11 Summary"), "2021-02-11
Summary")
})
Test passed

> test_that ("expect message", {
  expect_message(message("2021-02-11 Summary"), "Summary
Statistics")
})
— Failure (Line 2): expect message —————

`message("2021-02-11 Summary")` produced unexpected messages.
Expected match: Summary Statistics
Actual values:
* 2021-02-11 Summary
Error: Test failed

> test_that ("expect warning", {
  expect_warning(warning("Danger!"), "Danger")
})
Test passed

> test_that ("expect warning", {
  expect_warning(warning("Danger!"), "Danger Alert!")
})
— Failure (Line 2): expect warning —————

`warning("Danger!")` produced unexpected warnings.
Expected match: Danger Alert!
Actual values:
* Danger!
Error: Test failed

> test_that ("expect error", {
  expect_error(10 / quote, "Numeric argument required")
```

```
})
— Failure (Line 2): expect error —————

`10/quote` threw an error with unexpected message.
Expected match: "Numeric argument required"
Actual message: "non-numeric argument to binary operator"
Backtrace:
 1. testthat::expect_error(10/quote, "Numeric argument
required")
 2. testthat::quasi_capture(...)
 5. rlang::eval_bare(quo_get_expr(.quo), quo_get_env(.quo))
Error: Test failed

> test_that ("expect output", {
  expect_output(str(quote), "Science is curiosity, testing
and experimenting")
})
Test passed

> test_that ("expect output", {
  expect_output(str(quote), "A diagnostic message")
})
— Failure (Line 2): expect output —————

`str(quote)` does not match "A diagnostic message".
Actual value: " chr "Science is curiosity, testing and
experimenting""
Backtrace:
 1. testthat::expect_output(str(quote), "A diagnostic
message")
 2. testthat::expect_match(...)
 3. testthat::expect_match(...)
Error: Test failed
```

The *expect_lt*, *expect_lte*, *expect_gt*, and *expect_gte* functions compare two values to be less than or greater than and equal to each other. Examples for the same for the success and failure cases are illustrated in the following examples:

```
> test_that ("expect lt", {
  expect_lt(1, 2)
})
Test passed

> test_that ("expect lt", {
  expect_lt(2, 1)
})
— Failure (Line 2): expect lt —————

2 is not strictly less than 1. Difference: 1
Error: Test failed

> test_that ("expect lte", {
```